

Best Mobile App Penetration Testing Services in Norway

Mobile applications have become an important part of how organizations deliver banking, healthcare, retail, communication, financial, and business services. As applications increasingly handle sensitive information and connect to cloud-based APIs, identifying security weaknesses before attackers can exploit them is essential. [Best Mobile App Penetration Testing Services in Norway](#) help organizations assess the security of Android and iOS applications, identify vulnerabilities, and strengthen application defenses through structured security testing.

What Is Mobile App Penetration Testing?

Mobile application penetration testing is a controlled security assessment designed to identify weaknesses in an application's code, architecture, authentication mechanisms, data storage, communications, and interaction with the mobile operating system.

Unlike basic automated vulnerability scanning, penetration testing combines automated tools with manual analysis. Security testers examine how an application behaves under different conditions and attempt to identify security weaknesses that automated tools may not detect.

The assessment can cover both Android and iOS applications, along with the APIs and backend services that support them. OWASP's Mobile Application Security Testing Guide provides testing guidance for Android and iOS and includes techniques for areas such as authentication, network communication, cryptography, reverse engineering, and platform security. ([OWASP Developer Guide](#))

Why Mobile App Security Testing Matters

Mobile applications frequently process information such as customer details, authentication credentials, payment information, business data, location information, and personal records. If an application does not properly protect this information, attackers may be able to access sensitive data or abuse application functionality.

Mobile applications also operate in environments that organizations do not fully control. Users may install applications on personal devices, connect through public networks, lose their devices, or use outdated operating systems. Consequently, application-level security controls are important even when the organization's backend infrastructure has strong defenses.

Security testing provides developers and security teams with practical insight into how an application responds to potential attacks and where improvements may be required.

Key Areas Covered in Mobile App Penetration Testing

A comprehensive assessment can examine multiple components of a mobile application and its supporting infrastructure.

Authentication and Authorization

Testing can evaluate login mechanisms, password policies, multi-factor authentication, session management, account recovery, and authorization controls. Testers can also determine whether users can access functions or information beyond the permissions assigned to their accounts.

Secure Data Storage

Mobile applications may store credentials, tokens, personal information, configuration data, or other sensitive content locally. Testing can examine whether sensitive information is stored securely and whether appropriate platform security mechanisms are being used.

OWASP MASVS specifically includes secure storage and privacy among its mobile application security control areas. ([OWASP Mobile Application Security](#))

Network Communication

Applications regularly communicate with backend APIs and cloud services. Security testing can examine whether communications are properly protected, whether certificates are validated correctly, and whether sensitive information could be exposed during transmission.

Cryptography

Testing can review the use of encryption and cryptographic functions within the application. Weak algorithms, improper key management, hardcoded secrets, or inappropriate cryptographic implementations can create unnecessary security exposure.

API Security

Mobile applications commonly depend on APIs for authentication, data retrieval, transactions, and other business functions. Testing should therefore consider the application's backend endpoints as part of the overall attack surface. OWASP notes that remote endpoints are not covered by MASTG alone and may require complementary web and API security testing. ([OWASP Mobile Application Security](#))

Reverse Engineering and Tampering

Security testers may analyze application packages to understand how sensitive functionality is implemented. Testing can also evaluate whether an application has appropriate resilience against reverse engineering and tampering attempts. These areas are included within the OWASP MASVS control framework. ([OWASP Mobile Application Security](#))

Android and iOS Security Testing

Android and iOS have different architectures, security mechanisms, application packaging formats, and platform behaviors. A testing methodology should therefore account for the characteristics of the target operating system.

For Android applications, assessments may examine application permissions, local storage, exported components, intents, cryptographic implementations, and application package security.

For iOS applications, testing can examine application storage, keychain usage, inter-process communication, permissions, networking, authentication, and platform-specific security controls.

OWASP's MASTG provides separate guidance for Android and iOS testing, alongside general mobile security testing techniques. ([OWASP Mobile Application Security](#))

Following OWASP Mobile Security Standards

The OWASP Mobile Application Security Verification Standard (MASVS) provides a structured baseline for assessing mobile application security. Its control groups cover areas including storage, cryptography, authentication, network communication, platform interaction, code quality, resilience, and privacy. ([OWASP Mobile Application Security](#))

The MASTG complements MASVS by providing detailed testing procedures and test cases. Using these resources can help security teams establish a consistent methodology and ensure that important areas of the mobile application attack surface are considered. ([OWASP Developer Guide](#))

Organizations should treat these standards as a baseline and adapt testing to the application's architecture, business functionality, threat model, and risk profile.

Common Vulnerabilities Identified

Mobile application penetration testing may identify a wide range of security weaknesses, including:

- Weak authentication and session management
- Broken access controls
- Insecure local data storage
- Improper certificate validation
- Sensitive information exposure
- Hardcoded credentials or secrets
- Weak cryptographic implementations
- Insecure API endpoints
- Excessive application permissions
- Insecure inter-process communication
- Insufficient input validation
- Business-logic vulnerabilities
- Inadequate protection against reverse engineering
- Application tampering weaknesses
- Privacy-related security issues

The actual findings depend on the application's architecture, technology stack, functionality, and implementation.

Mobile API and Backend Testing

Testing only the mobile application package may not provide a complete picture of application security. Most modern mobile applications depend heavily on backend APIs for business operations.

For example, an application might appear secure on the device while its API incorrectly trusts client-side parameters or fails to enforce authorization on sensitive functions. An assessment should therefore consider the interaction between the mobile application and backend infrastructure.

Testing can evaluate authentication tokens, authorization controls, input validation, API responses, rate limiting, session handling, and business logic. Combining mobile application and API testing can provide a broader understanding of the application's security posture.

Benefits of Mobile App Penetration Testing

Organizations can gain several benefits from regular mobile security assessments.

Early vulnerability discovery: Security weaknesses can be identified before applications reach a wider user base.

Better protection of sensitive data: Testing can reveal weaknesses that could expose credentials, personal information, or business data.

Improved authentication: Assessments can identify weaknesses in login, session, authorization, and account-management functionality.

Stronger API security: Testing can uncover vulnerabilities within backend services supporting mobile applications.

Secure development improvements: Findings can provide developers with practical feedback for improving application architecture and code.

Risk prioritization: Security teams can prioritize remediation according to technical severity and business impact.

When Should a Mobile App Be Tested?

Mobile penetration testing can be performed at different stages of the software development lifecycle. Testing before a major release can help identify security issues before deployment.

Assessments may also be useful after significant application changes, backend modifications, authentication changes, new integrations, or major architecture updates.

Organizations with frequently updated applications can integrate security testing into their development and release processes. OWASP notes that MASVS controls can also support automated unit, integration, and acceptance testing as part of secure development practices. ([OWASP Mobile Application Security](#))

Choosing a Mobile Security Testing Provider

When selecting a testing provider, organizations should consider experience with both mobile platforms and application security. The provider should understand Android and iOS architectures, API security, authentication, secure storage, cryptography, and mobile-specific attack techniques.

A well-structured engagement should begin with clearly defined scope and objectives. The provider should explain the testing methodology, required access, testing environments, rules of engagement, reporting format, and remediation process.

The final report should provide enough technical evidence for developers and security teams to reproduce and understand findings. It should also include practical remediation guidance rather than simply listing vulnerabilities.

Building a Continuous Mobile Security Program

Mobile application security should not be treated as a one-time activity. Applications evolve continuously, and new features, APIs, third-party libraries, operating-system updates, and integrations can introduce new risks.

Organizations can combine penetration testing with secure code reviews, threat modeling, dependency management, automated security testing, API assessments, vulnerability management, and developer security training.

A continuous approach helps organizations identify security weaknesses earlier and incorporate security into the application's development lifecycle.

Strengthen Your Mobile Application Security

A structured mobile penetration testing program can provide valuable insight into application weaknesses and help organizations improve the protection of sensitive information. By combining manual testing, automated analysis, API assessment, platform-specific testing, and recognized security guidance, organizations can build more resilient mobile applications.

Organizations looking to strengthen their mobile application security can work with [cybivalue](#) to explore cybersecurity assessment services and develop a testing approach aligned with their application and business requirements.

Contact Information

For more information about mobile application penetration testing and cybersecurity services:

Website: <https://www.cybivalue.com/>

Email: contact@cybivalue.com

Phone: 9364783713